

Workshop No. 1

Kaggle Systems Engineering Analysis

Grupo Bimbo Inventory Demand

Juan Pablo Arismendi Sánchez
Juliana Camila Rincón Rojas
Samuel Casas Cantor

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1. Introduction

The competition we chose from the wide variety of competitions available in Kaggle is **Grupo Bimbo Inventory Demand**.

The primary reason for selecting Grupo Bimbo is the accessibility and abundance of publicly available information related to its operations, logistics, and market presence. As one of the largest bakery companies in the world, Bimbo offers a rich context for understanding how inventory demand is managed in a highly complex supply chain that spans multiple countries and product lines.

The ability to manage inventory efficiently is essential for companies in the food industry, particularly those dealing with perishable goods. Therefore, this analysis aims to investigate the variables and processes that influence Bimbo's inventory demand, with the objective of maximizing sales and minimizing returns of bakery goods. By modeling the relationship between inputs, system processes, and outputs, we aim to propose informed strategies that support operational efficiency and customer satisfaction while reducing waste.

2. Competition Overview

The Grupo Bimbo Inventory Demand competition, hosted on Kaggle, serves as the starting point for this systems analysis. The challenge asked participants to predict the weekly adjusted demand for baked goods, considering product sales and returns. This task aligns closely with real-world operational concerns, where accurate demand forecasting can improve production planning, reduce waste, and optimize distribution logistics.

2.1 Problem Context

Grupo Bimbo operates in a highly dynamic environment where product shelf-life is short, distribution routes are vast, and demand can fluctuate due to many factors including time, location, client behavior, and product popularity. Overstocking leads to waste and returns, while understocking leads to missed sales and customer dissatisfaction. The central challenge lies in balancing inventory with actual demand, a problem that is both operational and systemic in nature.

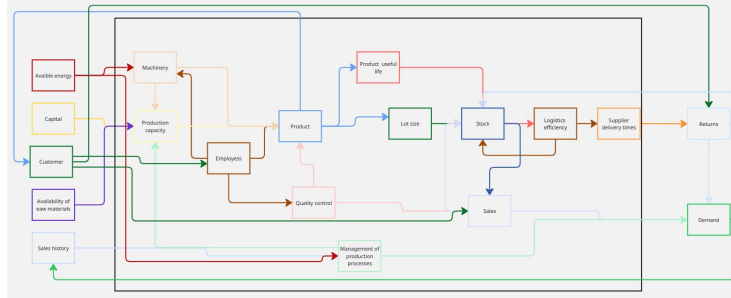
2.2 Dataset Overview

The dataset provided by the competition includes transactional records of product deliveries to clients overtime. It contains millions of rows of historical data, including details such as week number, sales location, client ID, product ID, and demand (in units). These variables serve as proxies for broader system inputs such as geography, customer behavior, product type, and seasonality.

While some variables in the dataset are more relevant to technical modeling (e.g., internal route codes), others directly support the system-level analysis — helping us understand how inputs like product type, client profile, and time periods interact to influence demand patterns and operational decisions.

3. Systemic Analysis

To optimize Grupo Bimbo’s inventory demand system, we analyze it as a dynamic process where various inputs are transformed into actionable outputs. The system’s goal is to ensure that the right amount of bakery products reaches the right clients at the right time, minimizing both missed sales (understocking) and returns (overstocking due to spoilage or unsold goods).



3.1 Input Variables (External Factors Driving the System)

These are external elements that influence demand and supply chain decisions:

- Available energy
- Capital
- Customer
- Availability of raw materials
- Sales history

3.2 System Variables (Transformation Processes within Grupo Bimbo's Operations)

These processes convert input into decisions that affect demand and returns:

- Machinery
- Production capacity
- Employees
- Product
- Quality control
- Management of production processes
- Product useful life
- Lot size
- Stock
- Sales
- Logistics efficiency
- Supplier delivery times

3.3 Output Variables (Performance Indicators & Business Outcomes)

These reflect the system's effectiveness:

- Returns
- Demand

4. Analysis

Now that we have a clearer understanding of the key processes involved in the system, we can delve deeper into the relationships between the various variables that determine the quality, efficiency, and overall success of the product reaching the consumer.

4.0.1 Product Quality Control (QC)

Quality Control (QC) serves as the first line of defense in ensuring that the product adheres to predefined standards of quality after the baking process. The primary goal of QC is to ensure that the product meets critical specifications, including appearance, consistency, packaging integrity, and freshness.

Inputs: Finished baked goods existing the production line.

Processes: Manual inspections, sensor-based evaluations, and adherence to standardized checklists.

Outputs: Accepted products are forwarded to storage, while rejected products are designated for disposal or reprocessing.

Dependencies and Implications:

- **Operator Attentiveness:** The accuracy of QC is heavily dependent on the attentiveness and skill of the operators conducting manual inspections. A lapse in attention or fatigue can result in incorrect assessments, potentially allowing substandard products to pass or rejecting products that would otherwise be acceptable.
- **Inconsistent QC:** If QC processes are inconsistent or poorly executed, it can lead to customer complaints, product returns, and increased operational costs due to unnecessary waste.
- **Stringent QC Thresholds:** While strict QC standards are vital for maintaining quality, setting excessively high thresholds can lead to an inflated rejection rate. This, in turn, increases waste and operational costs, potentially reducing overall efficiency.

4.0.2 Warehouse Storage and Batch Management

Once the products have passed QC, they are grouped into batches and stored in the warehouse until they are ready for distribution. Proper warehouse storage and batch management are crucial to preserving product integrity, managing inventory efficiently, and optimizing logistics.

Inputs: Approved batches from QC.

Processes: Labeling, tracking, environmental control (temperature and humidity), and batch rotation policies (FIFO/LIFO).

Outputs: Dispatched goods sent to distribution channels; flagged items due to shelf life constraints.

Dependencies and Implications:

- **Environmental Factors:** The preservation of product quality is highly sensitive to environmental conditions such as temperature and humidity. Any fluctuations outside the optimal range can lead to spoilage or deterioration of the product, thereby shortening its shelf life.
- **Batch Labeling and Misplacement Errors:** Errors in batch labeling or misplacement of products can create significant downstream issues, such as inaccurate inventory tracking, difficulty in locating products, or incorrect shipments.
- **Storage Duration:** The length of time products are stored directly correlates with waste and returns. The longer products remain in storage, the higher the risk of expiration, spoilage, or obsolescence, which could lead to customer dissatisfaction.

4.0.3 Shelf Life and Expiry Date Management

Since bakery products have inherently limited shelf lives, accurate management of shelf life and expiry dates is essential for minimizing waste, ensuring product quality, and optimizing inventory.

Inputs: Production timestamp, preservation conditions.

Processes: Shelf life estimation algorithms, expiry date assignment, and monitoring systems.

Outputs: Prioritized batch dispatch, discounted sales, or removal from inventory.

Dependencies and Implications:

- **Accurate Expiry Date Tracking:** Timely and accurate tracking of shelf life reduces the likelihood of expired goods reaching consumers. This not only protects the brand's reputation but also helps to avoid costly product returns.
- **Misestimation of Shelf Life:** Inaccurate shelf life predictions can lead to products being sold past their prime, resulting in financial losses due to returns, discounted sales, and customer dissatisfaction.
- **Real-Time Data Integration:** Real-time shelf life tracking must be integrated into the forecasting and logistics planning processes to optimize

dispatch schedules, prevent stockouts or overstocking, and improve overall supply chain efficiency.

4.0.4 Human Factor: Production Line Operators

The human factor plays an irreplaceable role in ensuring the smooth functioning of all processes, from QC to batch handling and storage. Operators serve as the bridge between automated systems and manual inspections, influencing the quality of the final product at every stage.

Inputs: Workload, training level, shift duration, ergonomic conditions.

Processes: Inspection tasks, machine handling, QC documentation.

Outputs: QC reports, incident logs, operational deviations.

Dependencies and Implications:

- **Fatigue, Stress, and Low Morale:** The well-being of production line operators is a critical factor in maintaining consistency and accuracy throughout the process. Fatigue, stress, and low morale can introduce errors in inspections, machine handling, or documentation, leading to inefficiencies and potentially compromised product quality.
- **Training and Operator Skill:** Well-trained, alert operators are essential to enhancing the robustness of the system. Proper training can reduce error rates, improve response times to operational issues, and ensure a higher level of quality control.
- **Human Oversight in QC:** Despite advancements in automation, human oversight remains essential for nuanced judgments during QC. Operators can detect subtle inconsistencies that machines may overlook, and their decision-making plays a crucial role in maintaining the overall quality of the product.

4.0.5 Product

The product is the outcome of all production activities. Its design, quality, cost, and availability are critical to the company's market success.

Inputs: Quality controls, employees, machinery, production capacity.

Processes: Assembles or manufactures the product according to specifications, applies quality control checks during and after production, packages and prepares the product for delivery or sale.

Outputs: Final goods ready for distribution or sale, product quality reports.

Dependencies and Implications:

- Well-designed, high-quality products improve competitiveness: when a product is well designed, it meets customer expectations and builds trust. This improves the firm's reputation and attracts new customers. As a result, demand for the product increases.
- Customization or innovation in products can increase market differentiation: Offering customized or innovative products allows the firm to meet specific customer needs and differentiate itself from competitors. This can create a unique value proposition and attract niche or premium markets.

4.0.6 Machinery

Machinery is a key component in any production system, directly affecting the speed, precision, and scale of output. Proper use and maintenance of machinery ensures efficient operations, consistent product quality, and reduced downtime.

Inputs: Available power, raw materials, trained operators.

Processes: Transforms raw materials into finished or semi-finished products, complete complex tasks with speed.

Outputs: Processed ingredients.

Dependencies and Implications:

- Efficient and modern machinery improves output: Modern machinery is designed to operate faster and with greater reliability than older equipment. This means the firm can produce more units in less time (increased output), with less errors or material losses.

4.0.7 Production capacity

Production capacity refers to the maximum number of products a firm can manufacture within a given time frame (in this case, a day), based on its available resources such as machinery, labor, raw materials, and employees. Understanding and managing production capacity is essential to meet demand.

Inputs: Available machinery and equipment, workforce availability and productivity, raw material supply, power availability.

Processes: Evaluates current resource levels to determine potential output, plans production schedules according to capacity limits.

Outputs: Capacity utilization rates, reports for scaling or optimization decisions.

Dependencies and Implications:

- **Strategic Capacity Planning Ensures the Firm Can Respond to Market Demand Fluctuations Effectively:** Strategic capacity planning involves forecasting demand and aligning production capabilities accordingly. By doing this, the firm can avoid being caught off guard by sudden increases or decreases in customer orders.

4.0.8 Production processes

Production processes are the core operations that transform raw materials into finished products. These include every step from initial preparation and assembly to quality control and packaging. The efficiency of these processes directly affects the firm's ability to fulfill demand and maintain product quality.

Inputs: Raw materials, available power, labor and machinery.

Processes: Converts raw materials into finished goods, applies standard operating procedures, coordinates with inventory and distribution teams for output alignment.

Outputs: Finished products, production performance data.

Dependencies and Implications:

- **Strong Production Processes Improve Product Consistency:** When production processes are well-designed and standardized, the firm can consistently produce items with the same quality, reducing variability and defects. This consistency builds customer trust and reduces rework or waste.
- **Real-Time Monitoring Enhances Adaptability to Changes in Demand or Resources:** Using real-time monitoring systems allows the firm to track production performance, resource usage, and output instantly. This enables quick decision-making when facing unexpected changes.

4.0.9 Raw material disponibility

Raw material availability is a basic element in any production process. The firm's ability to maintain continuous and efficient production depends on the

consistent supply of quality raw materials. Any disruption like transportation delays, or market shortages can affect the entire production chain.

Inputs: Supplier delivery schedules, inventory levels, high quality material.

Processes: Coordinates with suppliers to ensure timely deliveries, making sure quality materials.

Outputs: Any amount of quality raw material, updated inventory reports.

Dependencies and Implications:

- Strong dependency on external suppliers and logistics networks: The firm relies heavily on third-party suppliers and transport systems to deliver raw materials on time. Any delays, disruptions, or issues in these external networks can directly impact production schedules and overall efficiency.

4.0.10 Available Power

The availability of power is a critical operational factor in any production-based firm. Power directly impacts the functionality of machinery, production lines, refrigeration systems, and distribution processes. Without sufficient and reliable power, the firm risks delay and increased operational costs due to downtime or alternative energy sourcing.

Inputs: Energy consumption estimates, operational schedules.

Processes: Monitors energy usage across production and distribution areas, allocates power resources based on daily operational demands.

Outputs: Estimated time for operation, estimated time for productive processes.

Dependencies and Implications:

- High dependency on external energy providers or internal generators: The firm's operations are heavily reliant on a stable power source. This dependency means that any issues with external energy providers, such as blackouts, price spikes, or supply disruptions, can directly impact production. Similarly, if the firm uses internal generators as a backup or primary source, they must ensure regular maintenance, fuel supply, and efficiency.
- Efficient power management supports sustainability goals and cost efficiency: Effectively managing power consumption not only reduces unnecessary energy usage and operating costs but also contributes to broader sustainability objectives. This approach helps minimize waste, reduce energy bills, and align the company with environmental regulations or green

certification standards-enhancing both its public image and long-term viability.

4.0.11 The firm's capital

When discussing a firm, its operations are always intrinsically tied to its capital or budget. For the purposes of this analysis, it is crucial to focus solely on the portion of the capital allocated to specific areas: food production, product distribution, employee salaries, and the machinery required for production processes.

An essential factor in this analysis is determining the daily spending limit available for production. Understanding this will help maintain financial sustainability while maximizing productivity.

Inputs: Selling profit, production budget.

Processes: Determines the amount of capital to allocate for daily production activities, decides essential operational costs.

Outputs: Daily production capacity, updated capital allocation reports.

Dependencies and Implications:

- The firm's growth and operational stability: Effective capital management plays a crucial role in ensuring the firm's operational stability. Stability is essential, as disruptions in operations could lead to halted production or the inability to meet daily demand. Consistent capital flow allows the firm to maintain supply, satisfy customers, and support long-term growth.
- Any variation in selling profit affects production budget planning: Any fluctuations in selling profit significantly affect production budget planning. If profits fall below expectations, it can lead to serious financial constraints, reducing the funds available for daily operations, limiting production capacity, and potentially compromising the firm's ability to meet market demand.

4.0.12 Logistics Effectiveness

Logistics Effectiveness measures the efficiency and reliability of the supply chain in delivering products from production to customers. It encompasses transportation, warehousing, inventory management, and delivery timelines, directly impacting customer satisfaction and operational costs.

Inputs: Finished products, customer orders, transportation resources, warehouse capacity and location data, and supplier delivery schedules.

Processes: Route optimization minimizes delivery time/fuel costs; inventory allocation distributes stock by demand forecasts; real-time tracking monitors shipments via GPS; load balancing maximizes truck capacity; returns management handles reverse logistics.

Outputs: On-time conveyances, diminished transportation costs, minimized stockouts or overstock circumstances, client fulfillment measurements, information for nonstop change.

Dependencies and Implications:

- **Provider Unwavering quality:** The viability of coordination is affected by provider lead times and the consistency of conveyances. Delays or irregularities can disturb the supply chain, leading to stock deficiencies or an abundance of stock.
- **Foundation Restrictions:** Destitute street conditions, lacking warehousing offices, or transportation limitations can adversely affect coordination's execution, expanding costs and lead times.
- **Request Vacillations:** Sudden changes in client request can lead to overloading or understocking issues, influencing the productivity of coordination's arranging.
- **Innovation Integration:** Compelling coordination's depend on progressed following and administration frameworks. The need for integration between transportation systems, stock databases, and client request determining can lead to wasteful aspects and fumble.
- **Fetch Administration:** Adjusting cost-effective coordination's with opportune conveyances is vital. Cutting transportation costs as well forcefully can lead to delays, while prioritizing speed can expand operational costs.

4.0.13 Sales

Sales refers to the process of transferring inventory products for customers in exchange for payment. The main goal is to ensure light transactions, maximize income and maintain customer satisfaction while optimizing stock revenue and demand.

Inputs: Customer orders and purchase requests, product availability and inventory levels, pricing strategies and promotional campaigns, sales channels, and market demand and customer preferences, payments are handled through various methods, sold items are deducted from inventory in real-time.

Processes: Payments are handled through various methods, sold items are deducted from inventory in real-time, providing information, assistance, and personalized recommendations.

Outputs: Revenue from completed sales transactions, updated inventory reflecting sold products, sales reports and analytics for performance evaluation.

Dependencies and Implications:

- **Inventory Availability:** Sales depend on the stock levels; shortages lead to lost sales, while overstocking increases holding costs.
- **Market Demand:** Fluctuations in demand require adaptive sales strategies to avoid missed opportunities or unsold stock.
- **Payment Reliability:** Issues with payment processing can disrupt transactions, causing customer dissatisfaction.
- **Sales Channel Efficiency:** The effectiveness of retail stores, e-commerce, and direct sales impacts overall revenue generation.
- **Customer Satisfaction and Retention:** A smooth and positive sales experience encourages repeat business and strengthens brand loyalty.

4.0.14 Supplier Delivery Times

The delivery time of the supplier refers to an ordering time with a supplier and receiving raw goods or raw materials. This variable is essential to maintain a stable production and inventory level, ensuring that the materials are available if necessary to meet the production schedule and customer needs.

Inputs: Purchase orders sent to suppliers, supplier lead time estimates and historical delivery data, supplier reliability and capacity.

Process: Order processing and confirmation by the supplier, Preparation and packaging of goods by the supplier, transportation and shipment tracking.

Outputs: Raw materials or goods delivered to the production facility or warehouse, metrics on supplier performance.

Dependencies and Implications:

- **Supplier's reliability:** Coherent and accurate delivery time depends on the efficiency of the supplier and the reliability of operation. The delay or regular variant in the delivery schedule may break the production plan and stock management.
- **Transport and logistics:** External factors such as transport infrastructure, weather conditions and customs processes can affect delivery time, especially for international suppliers.
- **Dabbing shares:** Longer delivery time or unpredictable can require higher safety levels, which can increase maintenance costs and capital binding.
- **Production plan:** Delivery delays can stop or slow down production processes, affecting overall efficiency and capable of delaying the available products for sale.
- **Cost:** faster delivery options can lead to higher shipping costs, while the selection of shipping is slow and profitable and may be at risk of inventory deficiency.
- **Communication and coordination:** Effective coordinated and real-time communication with suppliers can help predict delays, adjust plans proactively and maintain the supply chain stability.

4.0.15 Client

Customers represent the last recipient of a product or service and are the main tools of the need in the system. Understanding customer needs, behaviors and satisfaction levels are essential to arrange production, inventory and logistics to meet the expectations of the market and maximize profits.

Inputs: Product offerings and availability, marketing and promotional information.

Process: Evaluation of product options based on preferences and needs, placement of orders or purchases through various sales channels, communication of feedback, complaints, or satisfaction levels, interaction with post-sale services.

Outputs: Purchase decisions and completed sales, customer feedback, reviews, and satisfaction data, market demand signals based on buying patterns, Requests for returns, exchanges, or support services.

Dependencies and Implications:

- Forecast requirements: The behavior of customers directly affects the request. The forecast is incorrect due to poor understanding of customers' trends that can cause excessive production or stocks.
- Customer satisfaction: A satisfied customer is more likely to become a repeated customer, while dissatisfaction can lead to complaints, output and negative oral transmission.
- Quality and the use of the product: customers expect the quality of reliable products and readiness quickly. The delay or defect can affect the reputation of the brand and the future sales.
- Sensitivity to price: Customers responded to price changes and promotions. Poor price strategies can reduce competitiveness or erode profits of beneficiaries.
- Market segment: different customer segments may have separate hobbies, requiring appropriate approaches in providing products, communication and distribution strategies.
- Managing relationships with customers: loyalty and long-term customer retention is affected by commitment after sale, personalization services and belief in the brand.

4.0.16 Sales History

Sales history refers to the recorded data of sales transactions made in a specified period. It serves as an important contribution to making decisions for different functions such as demand forecast, securities planning, marketing strategy and financial analysis. Sales history accurately and fully allows us to recognize models and analyze trends to optimize operations and better meet the needs of customers.

Inputs: Transaction records from point-of-sale (POS) systems or online platforms, customer purchase data (quantities, frequency, preferences), seasonal and promotional sales activity.

Process: Collection and storage of sales data in databases or ERP systems, data aggregation, sorting, and cleansing for analytical use, analysis of trends, seasonality, and customer purchasing behaviors.

Outputs: Reports and dashboards showing historical sales trends, inputs for demand forecasting, inventory planning, and procurement strategies, performance evaluations for products, regions, or time periods.

Dependencies and Implications:

- Data accuracy: The reliability of decisions based on sales history depends on the accuracy and complete properties of the recorded data. The input is missing or incorrect can deform the analysis and lead to wrong conclusions.
- Market conditions change sales history reflects past behaviors, not always predicting future demand, especially in the volatile market or evolution.
- Integrated with other systems: Sales history must be effectively integrated into stock systems, logistics and production to allow planning based on data and feedback in real time.
- Lag during response time: Historical data without inputs can lead to a difference to adapt to the market changes immediately, leading to loss of opportunities or ineffective.
- Calculating seasonal and event: Photos and historical dip must be explained correctly and distinguish between relapse seasonal trends and on-time events (for example, peak advertising).
- Security and data security: Sales history related to customer data must comply with data protection regulations. Malf management can lead to legal issues and the loss of customer confidence.

4.0.17 Returns

Productivity refers to the process of customer introduction after buying due to errors, dissatisfaction, control errors or other reasons. Effective management is essential to maintain customer satisfaction, control costs and minimize stocks and logistics disorders. Output can also provide valuable comments to improve quality and product service.

Inputs: Customer-initiated return requests, Reason codes (e.g., defective, wrong item, late delivery, buyer's remorse).

Process: Validation and authorization of return requests, inspection and categorization of returned products (restock, repair, disposal), updating inventory and financial records.

Outputs: Restocked, repaired, or discarded products, refunds or replacement items delivered to customers, return rate metrics and customer feedback reports, adjusted sales and inventory data.

Dependencies and Implications:

- Refund policy design: Refinery policies suitable for customers can improve satisfaction and loyalty, but pampered policies can increase operating costs and abuse.
- Product Quality: High profit level can point out the issues of quality, poor packaging or compensation between marketing and real product performance. Determining models can cause significant improvements in the product.
- Stock management: Production must be re-integrated into the inventory system with accuracy to avoid the difference in stocks and ensure that the units are available in the state.
- Impact of cost: output produces reverse logistics, labor to check and potential losses due to insensitive goods. Handling poor performance may significantly affect profits.
- Customer experience: The effectiveness and transparency of the return to enhance confidence and loyalty to the brand. The delay or complications can lead to negative exams and loss of future problems.
- Risk of fraud: Without appropriate control, output can be used by fraudulent complaints or re-abuse of policies, especially in high or frequent products.

4.0.18 Returns

The need to mention customers' wishes and desires to buy a specific product or service at a certain price and time. This is a basic tool in supply chain planning, affecting production, stocks, price strategies and marketing efforts. Understanding and predicting demand is essential to arrange effective resources and minimize costs while maximizing income.

Inputs: Market trends and consumer behavior data, historical sales records and seasonal patterns, promotional activities and advertising efforts, feedback from distributors, retailers, and sales teams.

Process: Demand forecasting using statistical models and market intelligence, segmentation of demand by product, region, or customer type, Scenario analysis for best, worst, and most-likely demand projections.

Outputs: Demand forecasts and predictive analytics reports, Input for production scheduling, inventory planning, and procurement, Strategic insights for pricing, product development, and marketing.

Dependencies and Implications:

- Forecasting forecast: Forecasting inaccurate demand may cause excessive production process (causing inventory and excess waste) or production below (leading to stocks and loss of sales).
- fluctuations of the market: External factors such as economic recession, global crisis or emerging rivals can quickly modify needs, which makes it difficult to predict the use of historical data.
- Customer behavior: Changes in consumer preferences, brand awareness or technology application can significantly change the demand models.
- Price elasticity: Demand may vary depending on the price change - too high and the sales may decrease, too low and the margin can erode or create unsustainable vertices according to demand.
- promotions and seasonal promotions: Special offers and peak seasonal can cause temporary excess in demand that must be predicted and planned.
- communication between parts: Application data must be shared effectively between sales, marketing, operations and logistics to allow the implementation of synchronous and reaction decisions.

4.1 Complexity and Interactions

Complexity refers to the degree to which a system is difficult to analyze, predict, or understand. It arises from the numerous interdependencies and interactions within the system. Unlike simple systems, complex systems display behaviors that cannot be easily deduced from their individual components.

Table 1: Relationships Between Variables

Source Variable	Dependent Variable	Nature of Relationship
Operators	QC Accuracy	Direct, human-influenced
QC Accuracy	Batch Acceptance	Filtering mechanism
Batch Acceptance	Warehouse Load	Volume-dependent
Storage Conditions	Shelf Life Preservation	Environmental feedback loop
Shelf Life	Distribution Priority	Time-sensitive
QC Rejections	Waste Generation	Direct correlation
Expiry Date Management	Sales Viability	Time-constraint-driven
Labeling Accuracy	Inventory Tracking Integrity	Critical to logistic synchronization

4.1.1 Multilayer Interdependence

One of the most defining features of this subsystem is the multilayer interdependence between its various processes. Each stage of the system relies on the accuracy and timing of the previous steps. These interdependencies are critical, as even minor errors at one stage can ripple through the entire system, compromising the quality and efficiency of the product.

- **QC Misses a Defect:** For example, if an operator’s attention wavers due to fatigue, a product may pass the Quality Control stage despite having defects. This faulty product then contaminates the batch, leading to issues in subsequent processes like batch labeling, storage, and eventual distribution.
- **Mislabeling in the Warehouse:** Similarly, if a batch is mislabelled during storage, it can cause significant downstream consequences. A mislabeled batch may disrupt expiry-driven prioritization, resulting in expired products being shipped before fresher items, ultimately affecting customer

satisfaction and increasing waste.

- **Shelf Life Dependency on Storage Conditions:** Shelf life is not solely dependent on the production date, but also on the conditions under which the product is stored. These conditions—temperature, humidity, and air quality—are influenced by the diligence of operators and the reliability of the storage equipment. Poor storage conditions can accelerate the degradation of products, reducing their shelf life even if they are freshly produced.

4.1.2 Dynamic Process Interaction

The processes within this subsystem do not operate in isolation; instead, they are dynamic and interact with one another in real-time. The system continuously adjusts based on feedback, creating a responsive and adaptive workflow.

- **Immediate Feedback from QC to Storage:** The results of the QC process feed directly into storage decisions. If products pass QC, they are sent for storage; if they fail, they are either reprocessed or discarded. This real-time feedback ensures that only products meeting quality standards move forward in the system.
- **Shelf-Life Tracking and Batch Routing:** Shelf-life management is equally dynamic. As products age, their expiration dates become increasingly important in determining their routing. Automated systems continuously update the status of batches, dynamically prioritizing products that are nearing expiry for faster distribution or discounting.
- **Temperature Fluctuations Trigger Alerts:** Environmental control systems constantly monitor the temperature and humidity in storage areas. If there are any unexpected fluctuations—perhaps due to equipment failure or external factors—alerts are triggered, influencing the distribution plans by prioritizing the products that are most at risk of being compromised.

4.1.3 Uncertainty and Variability

The system is also influenced by external uncertainties and variabilities that can disrupt the normal flow of processes. These uncertainties introduce challenges

in maintaining consistency and stability, requiring the system to be adaptable and resilient.

- **Ambient Weather Conditions:** Warehouse climate control systems are highly sensitive to external weather conditions. Sudden temperature spikes, high humidity, or prolonged periods of extreme weather can compromise the efficiency of climate control systems, which directly impacts the shelf life of stored products.
- **Staff Shortages Impacting QC Thoroughness:** Another source of variability is human resources. Unforeseen staff shortages, such as due to illness or absenteeism, can affect the thoroughness of the QC process. This reduces the reliability of the quality control results, potentially allowing defective products to pass inspection or leading to unnecessary rejections.
- **Production Overflows and Warehouse Pressure:** When production volumes exceed capacity, the pressure on warehouse systems intensifies. Overflows can lead to cramped storage conditions, where products may not be stored according to ideal conditions, affecting their longevity and increasing the risk of waste.

4.1.4 Feedback Loops and Delayed Effects

The subsystem's complexity is further exacerbated by feedback loops, where actions taken in one area can create ripple effects, often with delayed consequences. These feedback loops can either amplify problems or contribute to inefficiencies that compound over time.

- **Operator Training and QC Inaccuracies:** If operators are inadequately trained, they may make errors during the QC process. These errors can result in products being rejected unnecessarily or, conversely, accepted despite failing to meet standards. Over time, this increases the number of rejected products, which lowers batch integrity and drives up return rates, exacerbating inefficiencies in the system.
- **Overly Conservative Expiry Estimates:** If the system adopts overly conservative estimates of shelf life—aiming to avoid the risk of selling expired products—it can lead to the early disposal of goods that are still

edible and sellable. This results in higher operational costs due to increased waste and creates an artificial demand for overproduction, further straining the system.

- **Improper Storage Rotation:** A failure to rotate stock according to batch dates (e.g., improper FIFO or LIFO practices) can result in expired products remaining in storage undetected. This not only leads to customer dissatisfaction when expired goods reach the market but also damages the brand's reputation, affecting future sales.

4.1.5 Nonlinear Outcomes

In this highly interconnected system, small disruptions can have disproportionately large effects. Even seemingly minor errors or deviations from the optimal process can escalate into significant issues.

- **Mislabeled Batches:** A single mislabeled batch can set off a chain of events that results in misprioritized shipping, causing delays in deliveries. This not only increases return rates but also disrupts customer satisfaction, leading to increased operational costs and lost business.
- **Temperature Regulation Efficiency Loss:** A slight decline in temperature regulation efficiency can significantly reduce the shelf life of products, even if no visible product damage is apparent. Such a seemingly minor issue can halve the shelf life of the entire batch, affecting profitability and causing unnecessary waste without any obvious external signs of spoilage.

4.2 Sensitivity Analysis

A sensitivity analysis is used to determine how changes in input variables affect the outcomes of a system or model. It identifies which variables have the most influence on the results and helps evaluate the system's robustness under varying conditions. By systematically altering inputs one at a time (or in combinations), sensitivity analysis reveals potential risks, critical dependencies, and areas where uncertainty could significantly impact performance or decision-making.

4.2.1 Operator Fatigue and Alertness

Scenario: A shift in alertness due to extended hours or lack of breaks, leading to operator fatigue.

Impact: A 10% decrease in operator attention can cause a 20% increase in QC failures or overlooked defects. This increase in errors can result in defective products passing through the QC process undetected.

Ripple Effects:

- **Defective Products Enter Distribution Chain:** Once defective products bypass QC, they enter the distribution chain, eventually reaching consumers.
- **Consumer Complaints and Recalls:** Customers may file complaints about the product quality, leading to potential recalls, which not only affect brand reputation but also incur significant costs.
- **Financial and Operational Strain:** The company faces both financial losses from product returns and operational strain from having to manage the recall process, rework, and ensure that future QC processes are tighter.

The impact of operator fatigue underscores the importance of implementing regular breaks, limiting shift lengths, and introducing real-time fatigue monitoring systems to ensure QC accuracy is maintained throughout operations.

4.2.2 Environmental Storage Variability

Scenario: An increase of 3°C in warehouse temperature during peak summer months.

Impact: This minor temperature fluctuation can lead to a 15–20% reduction in the average shelf life of products. Even a small deviation from optimal storage conditions can accelerate spoilage, which significantly impacts product viability.

Ripple Effects:

- **Increased Returns Due to Early Spoilage:** Products that spoil faster due to poor storage conditions are more likely to be returned by customers, increasing the volume of returns and customer dissatisfaction.
- **Higher Waste Costs:** The company incurs increased costs from wasted products that can no longer be sold, either due to spoilage or expiry.

- **Resource Strain from Re-Stocking:** To mitigate the shortfall in available products, more resources are required to replenish inventory, including potentially higher production costs and additional shipping logistics.

This scenario highlights the importance of maintaining consistent warehouse climate control and investing in predictive climate control systems that can anticipate and adjust for temperature fluctuations in real-time.

4.2.3 Shelf Life Misestimation

Scenario: Underestimating the real shelf life of products by just one day.

Impact: Misestimating shelf life by a single day can lead to the premature disposal of viable products, resulting in approximately 3–5% loss of stock. This is particularly critical for products with limited shelf lives, such as bakery goods, where even slight miscalculations can have significant financial consequences.

Ripple Effects:

- **Inventory Shortage:** Premature disposal of products results in inventory shortages, meaning the company may need to increase production cycles to make up for lost stock. This adds pressure to the supply chain and increases production costs.
- **Increased Cost of Production Cycles:** To compensate for the loss in inventory, the company may need to run additional production cycles, which come with increased operational costs, including labor, energy, and raw materials.

Accurate shelf life estimation is crucial to minimize waste. This scenario also demonstrates the importance of continuous monitoring and real-time adjustments to shelf life estimates based on environmental factors and storage conditions.

4.2.4 Batch Mislabeling Rate

Scenario: A 1% mislabeling rate during the packaging process.

Impact: A small mislabeling error can lead to incorrect batch routing, causing early spoilage of certain batches or mismatches in distribution. Products may be shipped based on inaccurate expiry information, disrupting the flow of goods.

Ripple Effects:

- **Distribution Mismatches:** Mislabeling affects the prioritization of products, leading to products with shorter shelf life being distributed first, while fresher products may be left behind, potentially causing customer dissatisfaction and higher return rates.
- **Increase in Returns and Reclassification Workload:** As a result of mislabeling, up to 8–10% more products may be returned due to quality or expiry issues. Additionally, the reclassification and re-labeling workload increases, placing strain on warehouse and QC personnel.

Implementing better automation in labeling and integrating barcode scanning with real-time tracking systems can help reduce the risk of mislabeling and improve distribution accuracy.

4.2.5 QC Stringency Thresholds

Scenario: Tightening defect tolerance limits by 5% in QC.

Impact: By making QC thresholds stricter, the rejection rate increases by approximately 7%. This means that a larger proportion of products will be deemed unfit for sale, either due to minor defects or variations from the established standards.

Ripple Effects:

- **Higher Rejection Rates and Reprocessing Costs:** With more products rejected, the company incurs higher costs associated with reprocessing, which involves additional labor, time, and resources.
- **Temporary Waste Spike:** Initially, tightening QC limits could lead to a temporary increase in waste, as more products are discarded during QC checks.
- **Potential Long-Term Brand Gain:** While the immediate effect may be an increase in waste, tightening QC standards could improve the perceived quality of the brand over time, leading to stronger customer loyalty, fewer complaints, and potentially higher sales due to the increased trust in the product's quality.

This scenario demonstrates the delicate balance between maintaining high product quality and managing operational efficiency. While strict QC thresholds are essential for quality assurance, they must be calibrated carefully to avoid unnecessary waste or rework.

4.3 Chaos and Randomness Analysis

It refers to a deterministic but highly unpredictable behavior that arises due to extreme sensitivity to initial conditions. Despite being governed by precise mathematical equations, chaotic systems exhibit behaviors that appear random and disordered over time. Key properties of chaos include:

- **Non-linearity:** Chaotic systems are typically non-linear, meaning their behavior cannot be simply predicted by summing their individual parts.
- **Determinism:** Chaos is not randomness—it follows deterministic rules, but small differences in initial conditions can lead to drastically different outcomes.
- **Long-term unpredictability:** Even though the system is deterministic, its evolution is not predictable in the long run due to its sensitivity to initial conditions.
- **Complexity:** Chaotic behavior often appears disordered, making it difficult to understand and analyze.

4.3.1 Sources of Randomness

Randomness in this context refers to unpredictable deviations in input variables or process conditions that do not follow clear deterministic trends. These include:

- **Human Behavior:** Despite training and standard procedures, operator performance can vary due to health, mood, and fatigue—all of which introduce random fluctuations in QC accuracy.
- **Environmental Noise:** Variations in temperature, humidity, or dust accumulation within the warehouse can introduce noise in storage conditions, impacting shelf life and spoilage rates.
- **Machine Failures and Sensor Drift:** Sensors used for expiry tracking or environmental monitoring may produce irregular readings over time due to wear and tear or calibration shifts.

These random elements often interact with deterministic processes, compounding the difficulty in forecasting outcomes.

4.3.2 Emergent Chaos in Feedback Loops

Chaos, in contrast to mere randomness, describes deterministic systems that display **sensitive dependence on initial conditions**, where even minuscule differences can lead to divergent long-term outcomes. This is particularly relevant in the following feedback loops within the subsystem:

- **Inventory Pressure Loop:** A delay in dispatch caused by mislabeling can lead to batch buildup in the warehouse. This increases environmental stress on storage systems (e.g., refrigeration), which may reduce shelf life. As shelf life declines, more products are pulled prematurely from the system, increasing restocking demands and overloading QC checkpoints. The increased QC burden raises the probability of operator error, restarting the loop.
- **Training and Error Loop:** Initial operator mistakes increase the frequency of rejected batches. A high error rate draws managerial attention, often leading to retraining initiatives. However, during the training period, reduced workforce availability can heighten stress and further elevate error rates. The timing and intensity of the intervention can determine whether the loop dampens or spirals.

These loops exhibit behavior that can appear erratic and irreversible unless actively monitored and corrected, fulfilling the criteria for chaotic dynamics.

4.3.3 Stochastic-Deterministic Hybridity

The production subsystem is best understood as a stochastic-deterministic **hybrid**:

- **Deterministic Elements:** Shelf life formulas, expiry tracking algorithms, QC procedures.
- **Stochastic Influences:** Operator attention, micro-climate variability, batch routing anomalies.

This duality means that while outcomes are theoretically predictable, in practice they are subject to continuous random perturbation. As such, traditional deterministic models fall short unless augmented with probabilistic or chaotic systems theory.

4.3.4 Modeling and Mitigation Strategies

To manage chaos and randomness:

- **Agent-Based Modeling (ABM)** can simulate how thousands of interacting components (e.g., products, operators, sensors) behave under stochastic conditions.
- **Monte Carlo Simulations** provide insights into probabilistic shelf life loss under varied warehouse conditions.
- **Chaos Monitoring** using entropy-based algorithms or Lyapunov exponents can alert managers when the system approaches chaotic tipping points.

Operational practices should include:

- **Redundant QC stations** to absorb human error shocks.
- **Real-time alert systems** based on deviation thresholds in temperature and labeling accuracy.
- **Cross-functional rotation** for operators to reduce fatigue-induced variance.

5. Conclusion

5.1 Key Findings

The analysis reveals that the system under study exhibits characteristics of both chaotic and complex systems. While it operates under deterministic rules—for example, machinery processing raw materials into finished or semi-finished products always produces the same outcome given the same input—the system displays unpredictable behavior due to its sensitivity to initial conditions.

In this context, demand forecasting acts as a critical initial condition. The entire production process is built upon projected sales; therefore, any deviation between expected and actual demand can cause disruptions or inefficiencies.

Furthermore, the system's elements interact in non-linear ways, generating emergent patterns (such as quality control responses). If a product fails quality inspection, the process must be restarted or adjusted, reflecting the system's reactive and evolving nature. Additionally, randomness (from supply chain

variations to market shifts) amplifies unpredictability, especially when multiple external variables collide or interact simultaneously.

5.2 Potential Strengths of the System

Adaptability:

The system demonstrates the ability to self-organize and adapt to changing conditions, which is crucial in dynamic and competitive environments. Employees continuously monitor machinery and production processes, while distributors ensure that products reach customers in optimal condition. These human components act as adaptive variables, capable of adjusting workflows, resolving issues, and aligning operations with real-time demand to minimize waste and reduce production costs.

Emergence of Innovative Behaviors:

Due to the complex interactions among components, the system may produce creative or efficient responses to shifting market preferences. For example, if consumer trends suddenly shift toward organic products, the system—through coordinated changes in production and distribution—can adapt by developing new offerings that meet those demands.

Scalability:

Thanks to its decentralized structure and interaction-driven evolution, the system can scale effectively. Increasing production capacity may involve acquiring new machinery, hiring additional personnel, or expanding distribution networks. While this requires greater investment and strategic planning, the system's structure supports scalable growth without central bottlenecks.

Learning Potential:

With appropriate feedback mechanisms, the system could evolve and improve based on historical performance. For instance, analyzing past sales data, error rates, or customer feedback allows the system to adjust strategies, improve accuracy in forecasting, and refine operational processes over time.

5.3 Weaknesses of the System

Difficult to Control or Optimize:

Since the system's behavior arises from complex interactions among multiple variables, making precise, targeted adjustments can be difficult. Optimizing one component might inadvertently affect another, leading to unintended consequences.

Vulnerability to Random Disturbances:

Although adaptable, the system remains sensitive to unexpected external shocks, such as supply chain interruptions, market crashes, or energy outages. These disruptions can push the system into dysfunctional or unstable states, reducing efficiency or halting production.

High Computational Cost:

Accurately modeling or simulating the entire system requires significant computational resources due to its complexity. In practice, this makes it more practical to monitor real-time operational data rather than attempt to fully simulate the system. Decisions are often more effective when based on live performance metrics rather than abstract models.